Pipeline Hazards & Resolution

Q. Define Pipeline Hazards. Classify and explain the three main types of pipeline hazards (Structural, Data, and Control) and briefly mention methods to handle them.

> **Definition to Remember**

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1. Structural Hazards

Occurs when two instructions in the pipeline need the **same hardware resource** at the exact same time (Resource Conflict).

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port.

2. Data Hazards

Occurs due to **Data Dependency**, meaning an instruction needs the result of a previous instruction that hasn't finished executing yet.

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3. Control (Branch) Hazards

Occurs due to **Branch instructions** (Jump, If-Else). The pipeline fetches instructions sequentially. When a branch is encountered, the target address isn't known until the execute stage. Meanwhile, wrong instructions may have been fetched.

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> **Must-Write Points (for 10 marks)**

> 1. **

> **Quick Recall**

> `Pipe
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